EXAMPLE END OF LECTURE 19.

$Q \Rightarrow$ Volume Velocity

BEAT BOX SW.

$$Q = v \cdot A = 0.1\,m/s \cdot \pi\left(0.25\,m/2\right)^2 = 0.000\,m^3/s$$

$$W = \frac{\rho_0 c\, k^2}{8\pi}\left|\overline{Q_s}\right|^2 = \frac{(1.21)\cdot(343)\cdot\left(\frac{2\pi}{11.4}\right)^2 Q_s}{8\pi}$$

$$W = 1.2 \times 10^{-6} \text{ WATTS}$$

(NOMINAL; MUCH << 1W)

$$L_W = 10\log_{10}\left(\frac{W}{W_{ref}}\right); \quad W_{ref} = 1 \times 10^{-12}\,W$$

$$\left\{ L_W = 61 \text{ dB re: } 1\,pW. \right\}$$

MONOPOLE CHECKS: 1. SMALL
2. "BREATHING"

LECTURE 3 SLIDES - MONDAY, MARCH 31, 2025.

SLIDE 4

5

6

7

8

9

AVERAGE OF F1 & F2; MEAN.

10

11

SAME SCALE ACROSS SLIDES.

12.

13

d → SEPARATION DISTANCE.

14

15

16

17 - 21

22

GREEN'S FUNCTION.

$1/r$ ; DECAY

$\exp\{-jkr\}$ ; PHASE CHANGE.

$kd$ → SOURCE SIZE COMPARED TO WAVELENGTH

BREATHING → MONOPOLE
SUB-WOOFER

FAN BLADE → DIPOLE

ROTATING → LAT. QUADRUPOLE.